

NEURON

AI Opportunities for Property Platforms

Features, product value and their UK equivalents

21

property websites reviewed

10

highlighted AI features

3

priority groups

1

shared data and
intelligence layer

Feature priorities, development approaches and examples from around the world

Glossary: property platform – a website or app for finding, buying or renting property · equivalent – something that does the same job in another place
· layer – one level of a software system

What This Study Covers

Neuron is a multi-tenant platform built by Iceberg Digital. It serves 30+ UK estate agencies, and 21 of its websites were reviewed in this study.

The opportunities focus on four areas: making property search easier, making listing information easy to access, giving personalised suggestions, and supporting the decision process.

The highlighted features fall into three groups: core search and discovery, decision support, and advanced and special solutions.

Not all features bring the same product value or need the same technical effort, so they must be prioritised.

The aim is not to offer AI as a separate tool. The aim is to integrate it into the existing Nuxt 3 + API infrastructure.

Glossary: multi-tenant – one system that serves many separate customers · estate agency – UK word for a real estate office · Nuxt 3 – a framework for building websites · API – a way for software systems to talk to each other · prioritise – decide what to do first · integrate – combine into one system · infrastructure – the basic technical systems

Evaluation Criteria

01

User impact

02

Data readiness

03

Development
complexity

04

Commercial potential

Features were judged not only by how impressive they are as AI, but also by these four criteria.

GOAL

Support the whole flow with the same property and user data layer:

Search



Understand



Explore



Compare



Decide



Get in touch

OVERVIEW

10 Features, 3 Groups

Not all features bring the same product value or the same technical complexity.

CORE SEARCH AND DISCOVERY

Directly improves the search and listing discovery experience.

1 Natural-language search

2 Conversational search

3 Listing-specific Q&A

4 Area and neighbourhood guide

5 Personalised recommendations

It is at the centre of Phase 1 and Phase 2; this group builds the shared infrastructure.

DECISION SUPPORT

Makes it easier to choose between the properties that were found.

6 Property comparison

7 Match score

Low-to-medium development cost; it is added on top of the same data layer.

ADVANCED AND SPECIAL SOLUTIONS

Needs stronger data, image processing or a special user segment.

8 Visual and semantic search

9 Investment analysis

10 Guarded search assistant

Needs image processing and embedding infrastructure; it is positioned as a premium product.

Most of these features are not independent systems; they reuse the same AI and property data infrastructure.

Glossary: embedding – a list of numbers that represents the meaning of a text or image · infrastructure – the basic technical systems that everything else uses · segment – a specific group of customers · premium – higher-priced and higher-value · Q&A – questions and answers

Natural-Language Search

WHAT DOES IT DO?

Instead of choosing filters one by one, the user writes a single sentence:

"A 3-bedroom house with a garden, near a school, under £300,000."

The AI turns this sentence into search criteria such as location, price range, number of rooms, property type, garden, and transport or school distance. The result is sent to the existing search system. Unsupported or unclear criteria are clearly shown to the user.

HOW DOES IT ADD VALUE?

Users describe what they need in their own words, without learning how the filter system works. This layer is built once and can be used on many estate agency websites.

WHAT DO WE GAIN?

A shared criteria structure that all other AI features will be built on, and search data that shows what users really look for.

IS THERE A UK EXAMPLE?

Yes. Rightmove AI Search offers property search in natural language. Zoopla uses smart tags, made from text and image analysis, to add features to search that normal filters do not have.

Conversational Search

WHAT DOES IT DO?

Turns search from a single query into a two-way conversation:

*"3-bedroom house near Manchester" → "a bit cheaper" →
"maximum 20 minutes to a train station" → "remove the ones
without a garden."*

The system keeps the context of the earlier conversation, updates the active filters, and only calls allowed search functions.

HOW DOES IT ADD VALUE?

House hunting is not a one-time filter. People change their criteria as they see results. This feature makes that behaviour natural and keeps the user on the site.

WHAT DO WE GAIN?

A strong set of signals about the user's real preferences. It is direct input for the recommendation system and the match score.

IS THERE A UK EXAMPLE?

Yes. Ask Rightmove moves search from the classic filter system to a chat-based experience. In the US, Redfin Conversational Search and Zillow AI Mode use the same approach.

Glossary: context – earlier information that helps explain a new message · active filters – filters that are currently switched on · signal – a piece of data that shows what a user wants · recommendation system – software that suggests things a user may like

Listing-Specific Q&A (Ask This Property)

WHAT DOES IT DO?

On every listing page, users ask questions directly about that property:

"Does the driveway fit two cars?" · "What does this energy rating (EPC) mean?" · "Where is the nearest train station?"

Answers are based only on that listing's data: the description text, the energy certificate, the floor plan and the technical details. If there is not enough data, the system says so clearly and sends the user to the estate agent.

HOW DOES IT ADD VALUE?

The quality of listing descriptions differs a lot from agent to agent. Instead of searching a long text or asking the agent basic questions, the user gets the answer right on the listing.

WHAT DO WE GAIN?

Fewer repeated questions for agents, missing listing information becomes visible, and contact requests are of higher quality.

IS THERE A UK EXAMPLE?

Yes. Ask Rightmove extended its chat experience to the listing-page level and combines area data with listing information. In the US: Ask Redfin and Realtor.com RealAssist AI.

Glossary: EPC (Energy Performance Certificate) – an official document that shows how energy-efficient a building is · driveway – a private road between the street and a house · floor plan – a drawing of a property's rooms and layout · estate agent – a person who sells or rents property

Area and Neighbourhood Guide

WHAT DOES IT DO?

Describes not only the physical features of a property, but also the living conditions around it: schools, supermarkets, parks, stations, travel times, internet infrastructure and flood risk.

Instead of showing the data in a table, it explains what the data means:

"About 35 minutes to Liverpool Street; two supermarkets and three primary schools within walking distance."

Users can enter their own work or school address to get the travel time calculated.

HOW DOES IT ADD VALUE?

The user is really buying a combination of location and lifestyle, not just a building. Most of the reviewed sites only have a basic embedded map here.

WHAT DO WE GAIN?

Area content that competitors do not have, more time spent on the listing page, and knowledge of the user's real travel criteria.

IS THERE A UK EXAMPLE?

Yes. Ask Rightmove uses map and area data together with listing information. In the US, Zillow and Realtor.com RealAssist AI put transport and school criteria at the centre of search.

Personalised Listing Recommendations

WHAT DOES IT DO?

Builds a preference profile from user behaviour (viewed and saved listings, searches, price ranges, locations, skipped listings and contact requests) and scores candidate listings.

The first version can be rule-based, built on the similarity between listings. In later versions, embeddings, collaborative filtering and learning-to-rank methods are added.

HOW DOES IT ADD VALUE?

The "similar listings" areas on the reviewed sites are mostly not personalised; they only show the newest listings. A real recommendation system turns this area into value.

WHAT DO WE GAIN?

A relevance layer that can be measured through clicks and enquiry conversion, and more repeat visits.

IS THERE A UK EXAMPLE?

Yes. Zoopla reported more listing engagement and better lead results from AI-powered personalisation, and it also signed an enterprise agreement with OpenAI. In the US: Zillow recommendation systems.

Glossary: rule-based – works with fixed rules, not a learned model · embedding – numbers that represent the meaning of a text or image · collaborative filtering – suggesting what similar users liked · learning-to-rank – a method that learns the best order for results · conversion – a visitor taking a wanted action · lead – a possible customer · enterprise agreement – a large contract with a company

Property Comparison

WHAT DOES IT DO?

The user selects 2–4 listings. The system puts the data in the same format and produces a fixed comparison table. The AI does more than show the table: it explains the trade-offs.

"Listing A is £25,000 cheaper, but the journey to London is about 18 minutes longer. Listing B has a better energy rating and lower estimated energy costs."

HOW DOES IT ADD VALUE?

One of the hardest parts of house hunting is remembering and comparing listings in different browser tabs. This step can now be completed without leaving the site.

WHAT DO WE GAIN?

The decision stage stays inside the platform, the saved-listings system becomes active, and there is a shorter path to a contact request.

IS THERE A UK EXAMPLE?

There is no direct UK example, so this is a strong area for differentiation. In the US, Zillow AI Mode and Realtor.com RealAssist AI make comparison part of the chat experience.

Glossary: trade-off – a situation where you gain one thing but lose another · differentiation – being different from competitors in a useful way · estimated – roughly calculated, not exact · energy rating – a grade for a building's energy efficiency · journey – a trip from one place to another

Match Score

WHAT DOES IT DO?

This feature already exists in Neuron. Published properties that fit the buyer's or tenant's criteria (budget, number of rooms, area, garden/parking, etc.) are found, and each one gets a match score in percent. The reason for the score is also shown:

**Price 100% · Rooms 100% · Transport 75% · Garden 100%
· School proximity 60% → Overall match 87%**

Mandatory criteria and preferred criteria are separated, and the score is calculated with weighted scoring. If anyone has a suggestion for a new algorithm or a different approach, it is open to evaluation.

HOW DOES IT ADD VALUE?

Helps the most relevant listings stand out quickly among hundreds of results. Because the score is based on the user's real criteria, it is explainable and builds trust.

WHAT DO WE GAIN?

A low-cost, explainable ranking layer; over time, weights that learn from user behaviour.

IS THERE A UK EXAMPLE?

Yes, there is a close example. Rightmove AI Keyword Sort ranks results by checking keyword matches in listing images and text. Zoopla's smart tags infrastructure also works in this direction.

Glossary: weighted scoring – some criteria count more than others · explainable – it is possible to understand why a result was given · ranking – putting results in order · proximity – nearness · mandatory – required · algorithm – a set of steps that a computer follows

Visual and Semantic Search

WHAT DOES IT DO?

Users search by visual features that normal filters do not have:

"large windows", "modern kitchen", "brick wall", "Victorian style", "river view".

Listing photos are analysed with image-processing models. The visual features that are found are combined with the semantic features in the description text, and the result is written to the search index. The user's query is then matched with the listings that are closest in meaning.

HOW DOES IT ADD VALUE?

Not every important feature of a property is in the database. Much of it stays hidden in the photos and the description text.

WHAT DO WE GAIN?

Access to listings that competitors' filters cannot find, and a reusable property-feature dataset that enriches the whole portfolio.

IS THERE A UK EXAMPLE?

Yes. Zoopla's smart tags system extracts property features through text and image analysis, and Rightmove AI Keywords lets users search by features that normal filters do not have.

Investment Analysis

WHAT DOES IT DO?

For users who invest in property to rent it out, it produces an investment summary. The summary combines expected rental yield, return on investment (ROI), comparable sales, estimated renovation cost, rental demand, local market trend, energy rating and risks.

Financial calculations are done on the server with fixed formulas, not by the language model; the AI only explains the result. Data sources and assumptions are shown, and the summary states that it is not investment advice.

HOW DOES IT ADD VALUE?

It is not necessary for users who look for a standard home. But for investor-focused estate agencies, it is a strong differentiator and a way to qualify investment enquiries.

WHAT DO WE GAIN?

A product module for the high-value investor segment that can be packaged and priced separately.

IS THERE A UK EXAMPLE?

Yes. From a listing link, PropertyPortal.ai produces analyses of renovation cost, ROI, rental yield, energy rating and comparable sales. In the US: HouseCanary CanaryAI.

Glossary: rental yield – yearly rent as a percentage of the property price · ROI – profit compared with the money invested · comparable sales – recent sales of similar properties · renovation – repair and improvement work · assumption – something accepted as true without proof · qualify – check if an enquiry is serious

Guarded Search Assistant

WHAT DOES IT DO?

A search widget already works on the site. It has three modules:

- (1) It searches using only the existing filters in Neuron.
- (2) It sends users who want a valuation to the relevant pages.
- (3) It sends users who want to make contact to the contact-us page.

It deliberately does not go beyond these three modules, into unlimited free-text Q&A. The reason is explained in the box on the right.

WHY DO THESE LIMITS EXIST?

The questions that users can ask are unlimited, and AI can hallucinate.

For example, a user may describe their home and ask for a price. But commission differs from company to company, and a wrong price could cause serious problems.

For this reason, searches like "cheaper", "10 minutes to the metro" or "5 minutes to school" are not supported for now.

Glossary: guarded – protected by limits · widget – a small tool inside a web page · valuation – an expert estimate of what a property is worth · free-text – text that users write in any way they like · hallucinate – (AI) to produce false information that sounds true · commission – the fee an agent takes, usually a percentage of the sale price

Guarded Search Assistant: Our Suggestion

WHAT COULD OUR SUGGESTION BE?

Instead of letting the AI answer free text, we can apply structured-criteria conversion (function calling):

Location and transport phrases are converted into map data, and price phrases are converted into filters. The AI does not produce numbers.

On topics like commission, the assistant directs the user to the valuation or contact pages.

IS THERE A UK EXAMPLE?

Yes, the same principle is already in use.

Ask Rightmove and Redfin Conversational Search also do not answer free text directly. They only call allowed search functions and base the result on existing data.

The suggested approach follows this proven principle.

TECHNICAL VIEW

Shared Technical Architecture

Instead of building separate infrastructure for every feature, it makes more sense to build one shared data and intelligence layer.

Property database

Search and data services

Area and location data

User preference data

SHARED AI SERVICE LAYER

Natural-language and
conversational search

Listing-specific Q&A

Area and
neighbourhood guide

Recommendations
and match score

Comparison and
investment analysis

SHARED TECHNOLOGY SET

FastAPI / Node.js · PostgreSQL · Redis · Elasticsearch / OpenSearch · pgvector / Qdrant · OpenAI / Claude / Gemini · Google Maps and Places · existing property management and listing systems

Glossary: layer – one level of a system · FastAPI / Node.js – tools for building backend services · Redis – very fast temporary storage (cache) · Elasticsearch / OpenSearch – search engines for large amounts of data · pgvector / Qdrant – databases that store embeddings

PRODUCT PRIORITY OVERVIEW

Features Rated by Four Criteria

Feature	User impact	Data readiness	Development complexity	Commercial potential
1. Natural-language search	Very high	High	Medium	High
2. Conversational search	Very high	High	Medium	High
3. Listing-specific Q&A	Very high	High	Medium	High
4. Area and neighbourhood guide	High	Medium–High	Medium	High
5. Personalised recommendations	High	Medium	Medium	High
6. Property comparison	High	High	Low–Medium	Medium–High
7. Match score	Medium–High	High	Low–Medium	Medium
8. Visual and semantic search	Medium	Medium	High	Medium–High
9. Investment analysis	High for investors	Medium	High	High for specialist agencies
10. Guarded search assistant	High	High	Medium	High

Natural-language search, listing-specific Q&A and conversational search are the strongest starting points: besides being visible to users, they build the base for the other seven features.

Glossary: data readiness – how ready the data is for a feature · complexity – how difficult something is to build · commercial potential – the ability to make money · investor – a person who puts money into property to earn more · visible – easy to see

PHASE PLAN

Three Phases, One Shared Base

The goal is not ten independent features, but one product that grows on a shared data and intelligence infrastructure.

PHASE 1

Core and shared infrastructure

- Natural-language search
- Listing-specific Q&A
- Conversational search

The criteria structure, the property data index and the service layer built here form the base of all later features.

PHASE 2

Decision support

- Area and neighbourhood guide
- Match score
- Property comparison
- Personalised recommendations

Added on top of the same data layer; the extra infrastructure cost is low.

PHASE 3

Advanced and segment-focused

- Visual and semantic search
- Investment analysis
- Guarded search assistant

Needs image-processing models, embedding infrastructure and special datasets; it can be positioned as a premium product.

When Phase 1 is finished, most of the later features stop being a new-infrastructure problem and become a new-interface problem.

Glossary: segment-focused – built for one specific group of customers · criteria structure – the fixed list of search fields that the AI fills in · service layer – shared software that all features call · embedding – numbers that represent meaning · interface – the part of a product that users see and use

MARKET VIEW

UK Equivalents

Nine of the proposed features have a direct or very close example in the UK market.

Rightmove

Features 1 · 2 · 3 · 4 · 7 · 8 · 10

AI Search and Ask Rightmove (natural-language and conversational search, listing-level Q&A, map-based area data) · AI Keyword Sort (ranks results by analysing images and text) · AI Keywords.

Zoopla

Features 1 · 5 · 7 · 8

Smart tags created from text and image analysis: they find property features that classic filters do not have and add them to search · personalised search · an enterprise agreement with OpenAI; more listing engagement and better lead quality were reported.

PropertyPortal.ai

Feature 9

An investor-focused UK service that produces renovation cost, ROI, rental yield, energy rating and comparable-sales analyses from a listing link. It is the direct market equivalent of the investment analysis feature.

OPEN GAP

Property comparison has no direct UK example. In the US, Zillow AI Mode and Realtor.com RealAssist AI offer this experience. This gap is an early differentiation opportunity for Neuron.

Glossary: smart tags – labels created automatically from text and image analysis · enterprise agreement – a large contract with a company · lead – a possible customer · differentiation – being different from competitors · gap – an area that no one has filled yet

CONCLUSION

The global trend is to place AI directly inside property discovery and the decision process, not to write listing descriptions or add a chatbot to the site.

Search → Understand → Explore → Compare → Decide → Get in touch

This whole flow should be supported by services that use the same property and user data layer.

01

One shared layer

One shared data and intelligence infrastructure instead of 10 independent features; each new feature is added at a much lower cost.

02

Measurable product impact

More accurate results, more time spent on the listing page, and higher-quality contact requests.

03

Position in the market

Rightmove and Zoopla are moving in this direction; there are still gaps in comparison and match score.

Starting recommendation: natural-language search + listing-specific Q&A + conversational search

Glossary: trend – the general direction of change · measurable – can be counted or tracked with numbers · gap – an area that no one has filled yet · contact request – a message sent to an agent · position – your place in the market